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La position de la tête chez des personnes handicapées en chaise roulante peut-elle être associée à des malocclusions orthodontiques ?

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Division d'Orthodontie

Thèse préparée sous la direction des Professeurs Stavros KILIARIDIS et Gregory ANTONARAKIS

**"La position de la tête des personnes handicapées en chaise roulante
peut être associée à des malocclusions orthodontiques "**

Thèse
présentée à la Faculté de Médecine
de l'Université de Genève
pour obtenir le grade de Docteur en médecine
par

Kelly BILLIAERT
de
Lutry

Thèse n° 841

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**UNIVERSITÉ
DE GENÈVE**
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DOCTORAT EN MEDECINE DENTAIRE

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Intitulée :

La position de la tête chez des personnes handicapées en chaise roulante peut-elle être associée à des malocclusions orthodontiques ?

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Genève, le 6 juillet 2026

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Doyen

N.B. - La thèse doit porter la déclaration précédente et remplir les conditions énumérées dans les "Informations relatives à la présentation des thèses de doctorat à l'Université de Genève".

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INTRODUCTION

Les malocclusions dentaires constituent un problème de santé bucco-dentaire, dont la prévalence est significativement plus élevée chez les individus présentant des besoins spécifiques tels que handicaps physiques et/ou cognitifs[1, 2]. Plusieurs études épidémiologiques ont montré que moins de la moitié des personnes en situation de handicap présentent une occlusion considérée comme normale ou ne nécessitant pas d'intervention orthodontique[3]. Les béances antérieures, les articulés croisés postérieurs et les dysmorphoses sagittales sont particulièrement fréquentes dans cette population[4].

Cette forte prévalence souligne la nécessité d'identifier les facteurs étiologiques impliqués dans l'apparition et l'aggravation des malocclusions, afin de développer des stratégies préventives et thérapeutiques adaptées à ces patients.

Le développement des malocclusions est reconnu comme multifactoriel, résultant d'interactions complexes entre facteurs génétiques, environnementaux, neuromusculaires et fonctionnels[5]. Des altérations des fonctions oro-faciales, notamment, la respiration, la mastication, la déglutition et la posture linguale, sont fréquemment observées chez les personnes en situation de handicap et participent aux déséquilibres dento-squelettiques[6-9].

Dans ce cadre multifactoriel, la posture de la tête représente un paramètre fonctionnel susceptible d'influencer l'équilibre cranio-facial. Les relations squelettiques entre la tête, la colonne cervicale et le complexe maxillo-mandibulaire suggèrent qu'une modification prolongée de la posture de la tête pourrait affecter la position mandibulaire au repos, la direction de croissance faciale et, par conséquent, l'apparition de malocclusions.

LA POSITION NATURELLE DE LA TÊTE (Natural Head Position, NHP)

La position naturelle de la tête (NHP), mesurée quand « un homme est debout et que son axe visuel est horizontal, sa tête est dans une position naturelle »[10], est considérée en orthodontie comme une référence physiologique et reproductible pour l'analyse cranio-faciale. Les études céphalométriques ont montré des corrélations entre la position de la tête et certains schémas squelettiques, en particulier dans les dimensions sagittale et verticale. Une posture en extension a été associée à une augmentation de la

hauteur faciale antérieure et à des malocclusions de Classe II, tandis qu'une posture en flexion antérieure pourrait être liée à des malocclusions de Classe III[11, 12].

Cependant, ces observations reposent majoritairement sur des analyses bidimensionnelles statiques issues de photographies ou radiographies céphalométriques[13-17]. Ces méthodes ne permettent pas d'appréhender la dimension dynamique de la posture de la tête dans des conditions fonctionnelles quotidiennes. Cette limitation méthodologique justifie le développement d'outils d'évaluation tridimensionnels et dynamiques.

APPORT METHOLOGIQUE DES UNITES DE MESURE INERTIELLE (IMU)

Les unités de mesure inertielle (Inertial Measurement Units, IMU) constituent un outil technologique permettant l'analyse tridimensionnelle de l'orientation de la tête[18, 19]. Ces capteurs portables combinent généralement accéléromètres, gyroscopes et parfois magnétomètres afin de mesurer les rotations de la tête selon les trois axes principaux : pitch (flexion–extension), roll (inclinaison latérale) et yaw (rotation axiale)[20] (Figure 1).

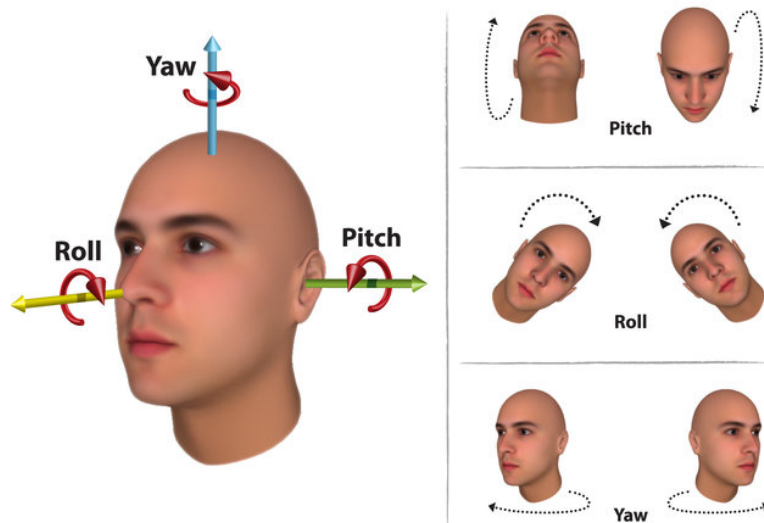


Figure 1 : représentation des 3 axes de mouvement de la tête

Arcoverde Neto et al., 2012

Grâce à leur capacité à enregistrer de manière continue les variations d'orientation dans l'espace, les IMU permettent une quantification objective et reproductible de la posture céphalique[21]. Leur utilisation présente également l'avantage d'être non invasive et sans exposition aux rayonnements ionisants, ce qui facilite leur application dans différents contextes cliniques et de recherche.

Leur format compact et leur facilité d'intégration dans des dispositifs portables permettent par ailleurs d'envisager leur utilisation dans l'étude de populations présentant des contraintes posturales spécifiques, notamment les individus utilisant un fauteuil roulant comme principal moyen de mobilité. Dans ce contexte, les IMU offrent la possibilité d'analyser la posture de la tête dans des conditions fonctionnelles proches de la posture habituelle des patients.

Ainsi, l'utilisation des unités de mesure inertielle ouvre de nouvelles perspectives pour l'étude objective et tridimensionnelle de la posture de la tête, en particulier dans des situations où les méthodes d'évaluation conventionnelles peuvent être plus difficiles à appliquer.

POSTURE DE LA TÊTE CHEZ LES INDIVIDUS EN FAUTEUIL ROULANT

Chez les individus utilisant un fauteuil roulant comme principal moyen de mobilité, la posture corporelle et l'alignement cranio-cervical peuvent être influencés par plusieurs facteurs, notamment le contrôle neuromusculaire, le tonus musculaire, l'installation posturale et l'utilisation éventuelle d'un appui-tête. Ces contraintes biomécaniques peuvent conduire à des adaptations posturales spécifiques, parfois maintenues de manière prolongée, susceptibles de modifier l'équilibre musculo-squelettique de la région cervicale et cranio-faciale.

Chez ces patients, la posture de la tête ne dépend pas uniquement des mécanismes physiologiques de stabilisation observés chez les individus sains, mais également de facteurs liés au contrôle postural global et à l'adaptation à la position assise prolongée. Une installation inadéquate peut favoriser l'apparition de positions de la tête compensatoires, influençant certaines fonctions telles que la respiration, la déglutition ou la communication[22, 23].

Malgré l'importance clinique de ces considérations, la posture de la tête chez les individus en fauteuil roulant reste relativement peu étudiée dans la littérature

orthodontique. La majorité des travaux portant sur la posture de la tête ont été réalisés chez des sujets sains, et les méthodes d'analyse utilisées reposent principalement sur des mesures réalisées en station debout[13].

Cette différence méthodologique soulève une question importante dans l'étude de la posture de la tête chez les personnes utilisant un fauteuil roulant. En effet, la position naturelle de la tête est traditionnellement évaluée en position debout, considérée comme la référence physiologique dans la plupart des études céphalométriques[24-26]. Cependant, chez les individus en fauteuil roulant, l'évaluation doit nécessairement être réalisée en position assise, qui correspond à leur position fonctionnelle habituelle.

Or, les différences biomécaniques entre les positions debout et assise pourraient influencer l'orientation de la tête et l'alignement cranio-cervical. Par conséquent, avant d'appliquer cette analyse à une population spécifique telle que les individus en fauteuil roulant, il apparaît nécessaire de s'assurer que l'évaluation de la posture de la tête en position assise constitue une méthode fiable et comparable aux mesures traditionnellement réalisées en station debout.

RELATIONS ENTRE POSTURE DE LA TÊTE ET MALOCCLUSIONS

Plusieurs études céphalométriques ont suggéré l'existence d'associations entre l'orientation céphalique et certains schémas de croissance squelettique, en particulier dans les dimensions sagittale et verticale. Ces observations ont conduit à l'hypothèse selon laquelle la posture de la tête pourrait influencer l'équilibre biomécanique du complexe cranio-facial.

La position de la tête est étroitement liée à l'alignement du rachis cervical et à l'équilibre des tissus mous cervico-faciaux. Des modifications prolongées de l'orientation de la tête pourraient ainsi entraîner des adaptations de la position mandibulaire au repos, de l'activité musculaire oro-faciale et de la direction des forces appliquées au complexe maxillo-mandibulaire. Dans cette perspective, certains auteurs ont suggéré qu'une posture en extension pourrait être associée à une rotation mandibulaire postérieure et à une augmentation de la dimension verticale antérieure, tandis qu'une posture en flexion pourrait favoriser une rotation mandibulaire antérieure[11, 12].

Ces interactions s'inscrivent dans le cadre des modèles fonctionnels du développement cranio-facial, selon lesquels la croissance des structures squelettiques est influencée par

l'environnement fonctionnel et neuromusculaire. Les adaptations posturales chroniques pourraient ainsi modifier l'équilibre musculaire et contribuer à l'expression de certains types de malocclusions.

Toutefois, les données disponibles dans la littérature restent hétérogènes. La majorité des études reposent sur des analyses céphalométriques bidimensionnelles et sur des observations réalisées dans des conditions statiques, ce qui limite la compréhension des interactions dynamiques entre posture de la tête et occlusion. De plus, peu de travaux ont exploré ces relations dans des populations présentant des contraintes posturales particulières, telles que les individus en fauteuil roulant.

Une meilleure caractérisation tridimensionnelle de la posture de la tête pourrait ainsi permettre d'approfondir l'analyse des interactions entre position de la tête et malocclusions dentaires. L'utilisation de méthodes de mesure dynamiques pourrait notamment contribuer à préciser le rôle potentiel de la posture dans l'expression des malocclusions.

OBJECTIFS DE LA THÈSE

Dans ce contexte, cette thèse vise à approfondir la compréhension des interactions entre posture de la tête et les malocclusions chez les individus en fauteuil roulant, en combinant validation méthodologique et application clinique.

Le premier article a pour objectif de valider l'utilisation des IMU pour l'évaluation de la position naturelle de la tête, en comparant les mesures obtenues en position assise et debout chez des individus sains.

Le second article a pour objectif d'évaluer la posture de la tête tridimensionnelle chez des individus en fauteuil roulant et d'analyser ses associations potentielles avec les caractéristiques occlusales.

En combinant ces deux approches, ce travail vise à contribuer à une meilleure compréhension du rôle fonctionnel de la posture céphalique dans l'expression des malocclusions et à proposer une approche diagnostique plus globale intégrant les dimensions posturales dans la prise en charge orthodontique des patients présentant des besoins spécifiques.

Measuring the difference in natural head position between the standing and sitting positions using an inertial measurement unit

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Abstract

Aims: The purpose of this study was to compare the natural head position (NHP) in the sitting position to the NHP in a standing position using inertial measurement unit (IMU) and lateral photographs.

Materials & Methods: Twenty healthy young adult volunteers were asked to look at a mirror located at 1 metre in front of their eyes while being recorded with the IMU system. Lateral photographs were also taken. This procedure was undertaken for the standing and sitting positions, on two separate occasions within a one-week interval.

Results: A strong correlation was found between the IMU system and the lateral photographs ($r > .99$) with regard to the pitch axis, the absolute mean difference was 0.4 ± 0.5 ($p = .99$) for both standing and sitting positions. We found that in the sitting position the head was elevated by 2.5 ± 2.4 ($p < .05$) more than in the standing position, but no significant differences were observed for the other two axes (roll and yaw).

Conclusion: The IMU system is comparable to lateral photographs for pitch assessment. Except for a slight elevation of the head in the sitting position, no clinical differences were observed for the NHP when comparing the standing and sitting positions.

KEYWORDS

clinicians, head, position, sitting position, standing position, wearable electronic devices

1 | INTRODUCTION

The concept of natural head position (NHP) was first introduced in the 19th century by Broca,¹ who defined it as the most balanced head position, when a person is standing and when their visual axis is horizontal. The deviations of NHP have been correlated to different alterations in several fields such as craniofacial morphology,² malocclusions,³ facial growth pattern,⁴ physiology of respiration^{5,6} and ophthalmology.^{7,8}

The analysis of head position is done on three axes (pitch, roll and yaw).⁹ The NHP is usually measured when a patient is standing in front of a mirror (namely the mirror-guided head position).¹⁰ However, analysing the NHP in a sitting position may be important for certain populations such as individuals with a handicap, the elderly or young children for whom it is not always possible to stand up. Few studies have focused on this subject. Bjerin¹¹ found that the head in the sitting position is slightly more elevated¹¹ and the variations in head balance were slightly greater.¹² However,

Billiaert and Al-Yassary authors contributed equally to this work.

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Huggare found no difference between the standing or sitting position, due to the methodology used, whereby the head position was corrected by the operator using a fluid level fixed on the head.¹³ Those previous studies focused only on the pitch axis; thus, it would be interesting to search if there are differences in the head position between the sitting and standing position looking at all three axes.

There are different methods available to measure the NHP such as the operator's estimation,^{14,15} lateral photographs¹⁶ or radiographs,¹⁰ head-mounted equipment¹⁷ and three-dimensional scans.¹⁸ The analysis can also be done with an electronic head-mounted equipment using the inertial motion unit (IMU)^{19,20} that can measure the three axes simultaneously. This system has already been tested in a previous study and showed excellent accuracy in measuring head position (M. Al-Yassary, K. Billiaert, G. S. Antonarakis, & S. Kiliaridis, submitted for publication).

The aims of this study were to:

1. Evaluate the NHP (mirror-guided head position) in all three axes comparing the standing and sitting positions;
2. Compare the observations on the pitch axis obtained from the electronic head-mounted equipment sensors to those acquired by the lateral photographs;
3. Analyse the reliability and variation of the measurements of the NHP for both the standing and sitting positions on two separate occasions.

2 | METHODS

2.1 | Ethics statement

The present study was approved by the Swiss Association of Research Ethics Committees. The experimental procedures were conducted in conformity with the Declaration of Helsinki. Informed consent for participation in the study and publication in an open

access format was obtained from the participants, with regard to their photographs and personal information. The procedures of the study were fully explained to the participants, and they provided their written informed consent before testing.

2.2 | Participants

Twenty healthy young adult volunteers at the University clinics of dental medicine in Geneva, Switzerland (7 men, 13 women), aged from 20 to 30 years, were recruited for this study.

2.3 | Experimental set-up and protocol

To perform a reproducible analysis, a standardised position is needed. For the sitting position, the 90–90–90 position described by Engström²¹ (corresponding to a 90° angle on the hips, the knees and the feet) is described as a reproducible and stable position. For the standing position, the orthoposition described by Molhave²² is that most commonly used for research purposes. We used two methods to analyse the NHP, namely lateral photographs allowing an assessment of pitch and wearable sensors (IMU) allowing an assessment of pitch as well as roll and yaw.

The participants were seated on a chair in the 90–90–90 position, facing a mirror which was one metre in front of their eyes. We calibrated the sensors on the ground to assess the neutral position (corresponding to 0° on each axis). The sensor was then placed on their forehead. We took a picture when the participants were gazing at their eyes in the mirror. Following this, the participants turned their heads to the photographer (for 5 s) then looked at the mirror for a second time (gazing at their eyes), when a second picture was taken. The same procedure was done with the participants standing up (Figure 1). In order to test the reliability of this method, the same procedure was performed on two separate occasions for each participant within a one-week interval.

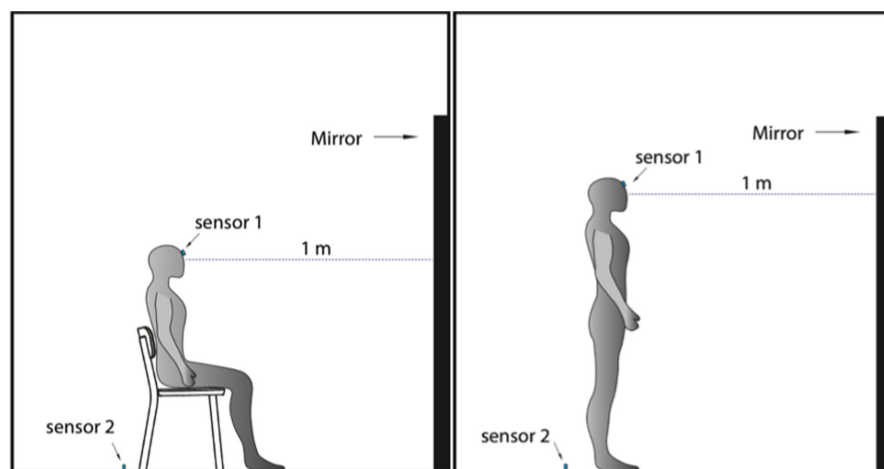


FIGURE 1 Representative figure of the set-up: The participants were seated on a chair in the 90–90–90 position 1 m in front of a mirror. After calibration of the inertial measurement unit on the floor to assess the reference, the first sensor was placed on the forehead to measure the natural head position. Following this, the participants stood up into a standing position at 1 m in front of the mirror and the natural head position was once again measured

2.4 | Lateral photographic measurement

We displayed a laser giving us the horizontal plane behind the participant. The lateral photograph allowed us to assess the pitch by measuring the angle between the Frankfurt plane and the horizontal plane given by the laser.

2.5 | IMU measurement

We used the MetaMotionR (MbleintlabInc., San Francisco, CA, USA) as the IMU, composed of two sensors. Before measuring the head position, this system needs to be calibrated. This is done by placing the first sensor on the ground in front of the participants and the second one behind them to assess the neutral position (corresponding to 0° on each axis). After calibration, the first sensor was placed on the forehead. Error of measurement could be caused by improper calibration, incorrect position of the sensor and the inclination of the forehead. To minimise the error caused by the calibration, we used the standardised procedure described above. To minimise the errors related to incorrect placement of the sensor on the forehead, we took a photograph in front of the participant to make sure that the sensor was parallel to the bipupillary line. Finally, for the possible sources of error in relation to forehead inclination, we took a lateral photograph and measured the inclination of the forehead. All angles are measured using Pixelstick (version 2.16.2, Plum Amazing Software LLC, USA).

2.6 | Statistical analysis

All statistical analyses were performed using SPSS (version 24.0, SPSS Inc., Chicago, IL, USA). For each participant, we took two lateral photographs and two IMU recordings for the sitting position and standing position respectively, and this was done on two separate occasions.

To analyse the difference between the standing and sitting positions, linear regression analyses were performed with the standing data as the independent variable and sitting data as the dependent variable. A Bland-Altman plot was used to visualise the spread of the differences between the two positions compared to their mean with a 95% confidence interval (95% CI).

To compare the pitch measured by the IMU system to the lateral photographs, Pearson correlation coefficients were calculated. Linear regression analysis was performed with the lateral photographic data as the independent variable and IMU data as the dependent variable. Intra-class correlation (ICC) (based on a single rating, absolute-agreement, 2-way random-effects model) estimates and their 95% CI were calculated.²³ ICC values less than 0.5 are indicative of poor reliability, those between 0.5 and 0.75 indicate moderate reliability, those between 0.75 and 0.9 indicate good reliability and those greater than 0.90 indicate excellent reliability.²⁴ The standard error measurement (SEM) of the difference

between standing and sitting was calculated for each ICC (using the formula $SEM = Standard\ Deviation \times \sqrt{1 - ICC}$).

We compared the two sessions, by calculating the paired t test for systematic differences between the two sessions. A Bland-Altman plot was used to visualise the spread of the differences between the two sessions compared with their mean with a 95% CI.

3 | RESULTS

3.1 | Evaluating the NHP (mirror-guided head position) in all three axes comparing the standing and sitting positions

When comparing the two positions, we found that in the sitting position the head was more elevated (corresponding to the pitch axis) by 2.5 ± 2.4 ($p < .05$). We found no significant difference in the roll and yaw axes. The Bland-Altman plot shows a bias line at -2.6° , with a 95% CI of -7.4 to $+2.6$. The data points are distributed equally below and above the bias line (Figure 2).

3.2 | Comparing the observations on the pitch axis obtained from the electronic head-mounted equipment sensors to those acquired by the lateral photographs

The absolute mean difference between the IMU and the lateral photographic data was 0.5 ± 0.5 for the pitch axis. A strong correlation was found between them ($r = .99$; $p < .05$). The ICCs were excellent ($ICC = 0.99$) with a 95% CI of 0.991 – 0.998 corresponding to an $SEM = 0.03^\circ$.

3.3 | Analysing the reliability of the measurements of the NHP for both the standing and sitting positions on two separate occasions

When comparing the two sessions, we observed no systematic differences for the three axes ($p > .05$) for the standing and sitting position. The correlation was strong for the pitch, moderate for the roll and poor for the yaw (Table 1). When comparing the differences between the two sessions, we observed that the differences in the sitting position were systematically lower than in the standing position for the roll axis ($p = .01$). No statistically significant differences were observed for the pitch and the yaw axes. (Table 1).

The Bland-Altman plot shows the bias line at 0.04° for the sitting position and 0.19° for the standing position. The data points are distributed equally below and above the bias line. Interindividual variation for the roll and the yaw were approximately $\pm 5^\circ$; however, for the pitch, the interindividual variation was higher (from 0° to -20°). (Figure 3).

FIGURE 2 Bland-Altman plot comparing the sitting to the standing positions: The x-axis represents the mean between the two and the y-axis the differences. Red lines indicate the 95% confidence interval, and the black line indicates the bias (mean of the differences). Red points indicate the inertial measurement unit system, while green points indicate the lateral photographs

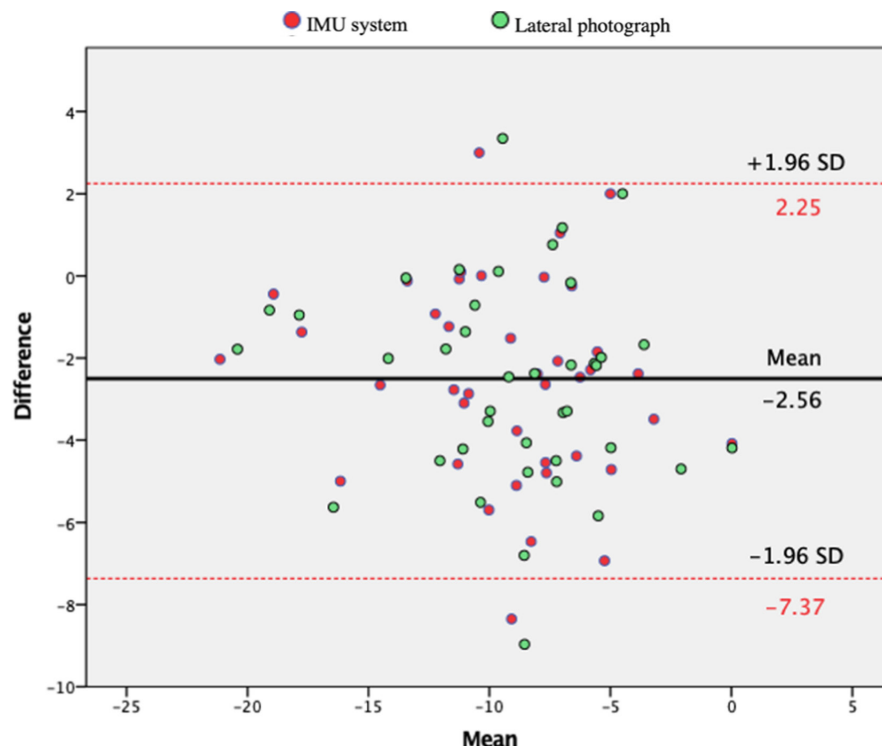


TABLE 1 Difference in natural head position between the two sessions (T^1 - T^2), using the inertial measurement unit system

	Sitting			Standing		
	Pitch	Roll	Yaw	Pitch	Roll	Yaw
Mean $\pm$ SD	-0.2 $\pm$ 2.1 ($p = .55$)	0.4 $\pm$ 1.4 ($p = .82$)	0 $\pm$ 2.3 ($p = .36$)	0.3 $\pm$ 2.5 ($p = .53$)	-0.4 $\pm$ 1.6 ($p = .75$)	0.7 $\pm$ 2.8 ($p = .78$)
r (p -value)	.89 (<.001)	.66 (<.001)	.27 (.25)	.87 (<.001)	.56 (.01)	.08 (.75)

Note: Means and standard deviations (SD) were calculated according to the differences between the first (corresponding to T^1) and the second (corresponding to T^2) sessions, done separately for each axis. The Pearson correlation (r) and p -value between the first and second sessions was also calculated.

4 | DISCUSSION

The present study, looking at differences in natural head position between the sitting and standing position, has shown that the head is slightly more elevated (pitch axis) in the sitting compared with the standing position. For the roll and yaw axis however, no differences were found when comparing the two positions. These results found on the pitch axis are similar to those found by Bjehin.¹¹

When comparing the measurement obtained by the electronic head-mounted equipment to the lateral photographs, we found excellent correlations. With a precise and adequate protocol, the electronic head-mounted equipment system can be used to measure the head position with great accuracy. The results found in our study were similar to the ones found by M. Al-Yassary, K. Billiaert, G. S. Antonarakis, and S. Kiliaridis (submitted for publication) measuring the head position, Beange et al.²⁵ measuring the movement of lumbar spine and Fennema et al.²⁶ measuring the flexion of the knee. This method is easy to use, quick to set up and can be used

on patients in the sitting or standing position. Moreover, the IMU system has an advantage over the other methods in that it allows an assessment of NHP not only in the pitch axis, but also in the roll and yaw axes.

The reliability of the measurements of the NHP on two separate occasions was different depending on the axis and the positions. The axis with the highest correlation was the pitch followed by the roll and yaw. With regard to the position, we found slightly greater correlation in the sitting position than the standing position in the three axes. The range of variations of the NHP on two separate occasions depends also on the axis and position. The axis with the least amount of variation was the roll, followed by the pitch and finally the yaw. Concerning the positions, we found no systematic difference in the range of variation for the pitch and the yaw. We did find however that the roll in the sitting position had systematically less variation compared to the standing positions. These results, in contrast to those found by Bjehin,¹¹ demonstrate that the sitting position is at least as reliable as the standing position and has the same range of variations or even less.

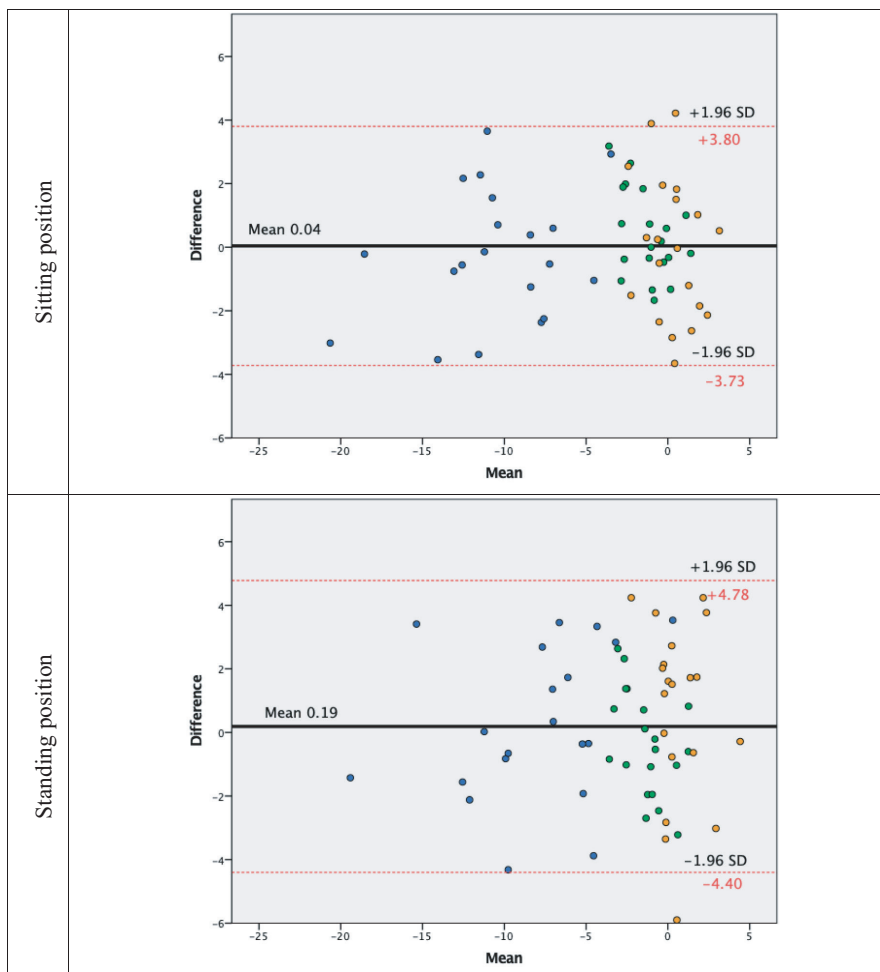


FIGURE 3 Bland-Altman plot showing the relationship between the two sessions: The x-axis represents the mean between the two and the y-axis the differences. Red lines indicate the 95% confidence interval, and the black line indicates the bias (mean of the differences). Blue points indicate the pitch, while green and orange points indicate the roll and yaw, respectively

The elevation of the NHP found when sitting compared with standing may also be present in patients who are generally seated. However, it is difficult to demonstrate this phenomenon due to the significant interindividual variation. The interindividual variation for the pitch is 4 times greater than for the roll or the yaw. This observation was also made by Huh et al.²⁷ who analysed the angle between the Frankfort plane and Sella-Nasion plane and found an individual variation ranging from 1.82° to 16.59°. If this elevation exists, it may have different implications on areas such as craniofacial growth and occlusion.

The results of the present study must be interpreted with caution because the participants included in our study were exclusively healthy young adults. The methodology used allows us to evaluate the short-term reproducibility. A one-week interval most likely prevents the participants from remembering the initial position, but it does not allow us to observe any changes in the head position over time. The methodology used allows us to evaluate the head position based on a one-shot procedure but the NHP is known to be a balance of several positions with a certain range of variations.¹⁴ It would be interesting to analyse the NHP for a long period of time in order to record the full range of variations and the tendencies for each axis.

5 | CONCLUSION

When comparing the NHP of the sitting to the standing position, no significant differences were found for the roll and yaw axes. For the pitch axis however, the NHP in the sitting position is slightly more elevated. The stability of the sitting position is comparable to the standing position for the pitch and yaw axes. However, the roll axis is more stable in the sitting position. The electronic head-mounted equipment, when used correctly, is comparable to lateral photographs for the pitch axis, with the advantage of recording the three axes simultaneously.

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CONFLICT OF INTEREST

The authors declare no competing interests.

AUTHOR CONTRIBUTIONS

K.B. and M.A. contributed to the data collection and data analysis; G.S.A. and S.K. contributed to the study design and drafting of the

manuscript. Figure 1 has been drawn by K.B., M.A. using Adobe Illustrator (version 24.3, cc 2020) (<https://www.adobe.com>)

PEER REVIEW

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DATA AVAILABILITY STATEMENT

All data included in this study are available upon reasonable request from the corresponding author.

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REFERENCES

1. Broca M. Sur les projections de la tête, et sur un nouveau procédé de céphalométrie. *Bull Soc Anthropol Paris*. 1862;3:514-544.
2. Solow B, Tallgren A. Head posture and craniofacial morphology. *Am J Phys Anthropol*. 1976;44:417-435. <https://doi.org/10.1002/ajpa.1330440306>
3. Solow B, Sonnesen L. Head posture and malocclusions. *Eur J Orthod*. 1998;20:685-693. <https://doi.org/10.1093/ejo/20.6.685>
4. Solow B, Siersbaek-Nielsen S. Growth changes in head posture related to craniofacial development. *Am J Orthod*. 1986;89:132-140. [https://doi.org/10.1016/0002-9416\(86\)90089-8](https://doi.org/10.1016/0002-9416(86)90089-8)
5. Cuccia AM, Lotti M, Caradonna D. Oral breathing and head posture. *Angle Orthod*. 2008;78:77-82. <https://doi.org/10.2319/011507-18.1>
6. Neiva PD, Kirkwood RN, Mendes PL, Zabjek K, Becker HG, Mathur S. Postural disorders in mouth breathing children: a systematic review. *Braz J Phys Ther*. 2018;22:7-19. <https://doi.org/10.1016/j.bjpt.2017.06.011>
7. Nucci P, Curiel B, Lembo A, Serafino M. Anomalous head posture related to visual problems. *Int Ophthalmol*. 2015;35:241-248. <https://doi.org/10.1007/s10792-014-9943-7>
8. Nucci P, Kushner BJ, Serafino M, Orzalesi N. A multi-disciplinary study of the ocular, orthopedic, and neurologic causes of abnormal head postures in children. *Am J Ophthalmol*. 2005;140:65-68. <https://doi.org/10.1016/j.ajo.2005.01.037>
9. Leung MY, Lo J, Leung YY. Accuracy of different modalities to record natural head position in 3 dimensions: a systematic review. *J Oral Maxillofac Surg*. 2016;74:2261-2284. <https://doi.org/10.1016/j.joms.2016.04.022>
10. Solow B, Tallgren A. Natural head position in standing subjects. *Acta Odontol Scand*. 1971;29:591-607. <https://doi.org/10.3109/00016357109026337>
11. Bjeih R. A Comparison between the Frankfort horizontal and the Sella Turcica -Nasion as reference planes in cephalometric analysis. *Acta Odontol Scand*. 1957;15:1-12. <https://doi.org/10.3109/00016355709041090>
12. Moorrees CFA, Kean MR. Natural head position, a basic consideration in the interpretation of cephalometric radiographs. *Am J Phys Anthropol*. 1958;16:213-234. <https://doi.org/10.1002/ajpa.1330160206>
13. Huggare JA. A natural head position technique for radiographic cephalometry. *Dentomaxillofac Radiol*. 1993;22:74-76. <https://doi.org/10.1259/dmfr.22.2.8375558>
14. Lundstrom A, Forsberg CM, Westergren H, Lundstrom F. A comparison between estimated and registered natural head posture. *Eur J Orthod*. 1991;13:59-64. <https://doi.org/10.1093/ejo/13.1.59>
15. Bass NM. The aesthetic analysis of the face. *Eur J Orthod*. 1991;13:343-350. <https://doi.org/10.1093/ejo/13.5.343>
16. Lundstrom F, Lundstrom A. Natural head position as a basis for cephalometric analysis. *Am J Orthod Dentofacial Orthop*. 1992;101:244-247. [https://doi.org/10.1016/0889-5406\(92\)70093-P](https://doi.org/10.1016/0889-5406(92)70093-P)
17. Audette I, Dumas JP, Cote JN, De Serres SJ. Validity and between-day reliability of the cervical range of motion (CROM) device. *J Orthop Sports Phys Ther*. 2010;40:318-323. <https://doi.org/10.2519/jospt.2010.3180>
18. Cassi D, De Biase C, Tonni I, Gandolfini M, Di Blasio A, Piancino MG. Natural position of the head: review of two-dimensional and three-dimensional methods of recording. *Br J Oral Maxillofac Surg*. 2016;54:233-240. <https://doi.org/10.1016/j.bjoms.2016.01.025>
19. Patel S, Park H, Bonato P, Chan L, Rodgers M. A review of wearable sensors and systems with application in rehabilitation. *J Neuroeng Rehabil*. 2012;9:21. <https://doi.org/10.1186/1743-0003-9-21>
20. Ghislieri M, Gastaldi L, Pastorelli S, Tadano S, Agostini V. Wearable inertial sensors to assess standing balance: a systematic review. *Sensors (Basel)*. 2019;19(19):1-25. <https://doi.org/10.3390/s19194075>
21. Engström B. *Ergonomic seating: a true challenge: seating and mobility for the physically challenged: risks & possibilities when using wheelchairs*. Stockholm: Posturalis Books; 2002.
22. Molhave A. The erect position in man; theoretical and statistical aspects: a biostatic study. *Nord Med*. 1959;62:1148-1150.
23. Portney LG, Watkins MP. *Foundations of Clinical Research: Applications to Practice*. Upper Saddle River, NJ: Pearson/Prentice Hall; 2015.
24. Koo TK, Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med*. 2016;15:155-163. <https://doi.org/10.1016/j.jcm.2016.02.012>
25. Beange KHE, Chan ADC, Beaudette SM, Graham RB. Concurrent validity of a wearable IMU for objective assessments of functional movement quality and control of the lumbar spine. *J Biomech*. 2019;97:109356. <https://doi.org/10.1016/j.jbiomech.2019.109356>
26. Fennema MC, Bloomfield RA, Lanting BA, Birmingham TB, Teeter MG. Repeatability of measuring knee flexion angles with wearable inertial sensors. *Knee*. 2019;26:97-105. <https://doi.org/10.1016/j.knee.2018.11.002>
27. Huh YJ, Huh K-H, Kim H-K, et al. Constancy of the angle between the Frankfort horizontal plane and the sella-nasion line: a nine-year longitudinal study. *Angle Orthod*. 2014;84:286-291. <https://doi.org/10.2319/062013-464.1>

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RESEARCH ARTICLE OPEN ACCESS

Evaluation of Head Posture in Wheelchair-Bound Individuals and Its Relation to Malocclusion: A Proof-of-Concept Study

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ABSTRACT

Background: Malocclusions are significantly more prevalent among individuals with physical and mental disabilities. Head posture, particularly in wheelchair-bound individuals, may play a role in craniofacial growth and occlusal development, but its specific impact remains poorly understood.

Objective: To evaluate head posture in wheelchair-bound individuals and assess potential correlations with malocclusion characteristics.

Methods: Twenty wheelchair-bound participants (aged 6–30 years) were categorised into three groups based on their head control (full, partial, or no control). Head posture was dynamically measured on three axes (pitch, roll and yaw) using an Inertial Measurement Unit for precise 3D tracking. Malocclusions, including molar and canine sagittal relationships, overjet, overbite and crossbites, were assessed through standardised clinical examinations.

Results: All wheelchair-bound participants showed significantly greater deviations in head posture compared to controls. The roll axis differed significantly across groups ($p = 0.004$), with the no control group exhibiting the greatest deviation. A significant correlation was observed between pitch axis deviation and sagittal molar relationships ($r = -0.59$, $p = 0.006$), indicating that a backward head posture was associated with Class II malocclusion, while a forward head posture was linked to Class III malocclusion. No significant correlations were found for the roll or yaw axes.

Conclusion: Significant head posture deviations were detected in wheelchair-bound individuals, with those having no head control showing more extreme deviations. This study found significant correlations between antero-posterior head posture deviations and sagittal molar relationships in wheelchair-bound individuals, suggesting that head posture could be a contributing factor to malocclusions in this population. Given the limited sample, these results should be interpreted cautiously.

1 | Introduction

Individuals with physical or mental disabilities exhibit a significantly higher prevalence of malocclusions [1]. Previous studies have reported that less than half of individuals with disabilities present normal occlusion or minor malocclusions that do not

require treatment [2]. Among this population, open bite, posterior crossbite and anterior crossbite are the most common types of malocclusions [3]. Individuals with intellectual disabilities are particularly at risk of developing malocclusions, partly due to altered oral functions such as mastication, swallowing and tongue posture [4, 5].

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The development of malocclusions is multifactorial, involving genetic, neuromuscular and environmental influences including compromised oral functions. Mouth breathing related to airway obstruction can alter tongue position and orofacial muscle function [6, 7], while parafunctional habits such as bruxism and thumb sucking, more frequent in this population, further contribute to the development of malocclusions [8, 9]. Additionally, genetic conditions associated with disabilities frequently involve craniofacial anomalies and malocclusion [10].

Head posture has been related to the development of malocclusions; research on craniofacial development based on lateral cephalometric radiographs suggests that head positioning can affect vertical growth patterns. Individuals who maintain a hyperextended head posture, such as in patients with obstructive sleep apnoea, or those who are blind, often exhibit a long facial structure, marked by greater anterior facial height growth and diminished posterior facial height growth [11, 12]. Despite these findings, few studies have examined head posture in individuals who rely on wheelchairs for daily mobility, a group in which head positioning is often constrained by seating configurations and neuromuscular control.

The elevated prevalence of malocclusion and limited access to orthodontic care among individuals with disabilities [4] underscore a substantial need for orthodontic treatment, including appropriate preventive measures in order to prevent or at least reduce the severity of the malocclusions witnessed. When considering wheelchair-bound individuals, anecdotal observations bring to light the potential problem of head posture, especially with inadequately adapted headrests. Wheelchairs and headrests much be customised for each patient, according to their unique needs. Improper head positioning may not only be related to the headrest itself but can also stem from the body's alignment in the wheelchair. Inadequate head support may lead to compensatory postures that disrupt oral functions such as mastication, swallowing, breathing and speech potentially influencing occlusal development. Proper customization of wheelchairs and headrests is therefore essential, as misalignment can exacerbate both postural and orofacial dysfunctions.

Given the multifactorial aetiology of malocclusions in this population, the present proof-of-concept study focuses on head posture as a potential contributing factor. The aim of this cross-sectional study was to evaluate three-dimensional head posture in wheelchair-bound individuals with varying levels of head control and to assess its possible association with malocclusion characteristics. As a proof-of-concept study, by comparing individuals with varying degrees of head control, the present investigation sought to determine whether impaired head posture may be a contributing factor in the development of malocclusions.

2 | Material and Method

2.1 | Study Design

The present cross-sectional study was approved by the local research ethics committee (CCER 2019-00659) and conducted in accordance with the Declaration of Helsinki. The study was

designed and reported with the STROBE guidelines [13] in mind to ensure methodological rigour, transparency and completeness in reporting. Given the exploratory nature of the present study, it was conceived as a proof-of-concept investigation aimed at generating preliminary evidence.

2.2 | Setting

This study was conducted in various institutions for individuals with disabilities across the French-speaking region of Switzerland (Swiss Romandy). Data were collected during scheduled visits to these institutions over a three-year period (from 2020 to 2023). The participating institutions included both residential facilities (specialised care homes) and day-care structures (specialised day facilities), representing a heterogeneous population with diverse medical, neuromuscular and syndromic conditions requiring wheelchair use. All visits were carried out following a standardised protocol to ensure consistency in data collection across all sites.

2.3 | Participants

Individuals were eligible for inclusion in the study if they were aged 6 years or older, in the early mixed, late mixed, or permanent dentition and had grown up using a wheelchair as their primary means of mobility. All eligible individuals present in the participating institutions during the study period were invited to take part. Informed consent was obtained from participants or their legal guardians, depending on age and cognitive ability.

Participants were categorised into three distinct groups based on their degree of head posture control. The first group comprised wheelchair-bound individuals demonstrating full control of their head posture, not requiring the support of a headrest (full head control group). The second group comprised wheelchair-bound individuals with partially limited head control who had headrests and used them when needed (partial head control group), while the third group was made up of wheelchair-bound individuals with no voluntary head posture control, fully dependent on a headrest for head positioning (no head control group).

A previously collected control dataset consisting of 20 healthy non-wheelchair-bound young adults was used for comparison [14]. These control measurements were obtained with the same IMU system, calibration protocol and testing environment, ensuring methodological equivalence.

The inclusion of this control group allowed deviations in head posture to be expressed relative to established normative values, improving interpretability.

The division of the sample into three distinct groups aimed to assess whether head position control achieved spontaneously or provided by the headrest (partially or completely) influences dentoalveolar growth and potentially contributes to the development of malocclusions.

Only a modest sample size was aimed for, reflecting the proof-of-concept nature of the study and the limited

eligible population available in the participating institutions. Although no power calculation was performed due to this study approach, this limitation is acknowledged and the results interpreted cautiously.

The data collected from the wheelchair-bound individuals (and from their parents or caretakers) were derived from a questionnaire, a clinical examination and an objective head posture assessment.

2.4 | Questionnaire

As part of the visits to the various institutions, a questionnaire was developed to gather key information about the participants. The questionnaire focuses on the individual's medical history, including the reasons for their use of a wheelchair.

Additionally, the questionnaire included questions regarding the participant's habitual posture in the wheelchair, including the position of their head. Lastly, questions concerning their dietary habits and oral hygiene practices were incorporated.

2.5 | Clinical Examination

A clinical intraoral examination was performed using an intraoral mirror and a periodontal probe, with appropriate lighting, allowing for a comprehensive assessment of intermaxillary occlusal relationships and the presence of malocclusions. Clinical assessment included sagittal molar relationships (Angle classification), overjet, overbite, and the presence any anterior and/or posterior crossbites. Additionally, an assessment of the general dental condition, including the detection of carious lesions and signs of periodontal disease was carried out.

2.6 | Head Position Assessment

Inertial measurement units (IMUs) were utilised to measure head position in three dimensions (pitch, roll and yaw). IMUs are electronic devices consisting of components such as a gyroscope, an accelerometer and a sensor that measures angular rate and gravity, that enable precise tracking of movements over a certain period of time and were chosen for their ease of use and accuracy [14–16]. With its use, a dynamic recording of head position over extended periods of time is achieved, making it possible to analyse the full range of natural head posture (NHP) variations during the recording.

The IMU system that was used for measuring head position was the Xsens Movella DOT Motion Tracking system (Movella Technologies B.V., Enschede, Netherlands), a wearable sensor developed for human kinematic analysis. Before starting the measurements, the IMU was calibrated to define the neutral position (set to 0° on all axes). This process involved placing the sensor on the ground in front of the participant, ensuring it was perpendicular to the ground. After appropriate calibration, the sensor was moved to the participant's forehead for the actual dynamic head posture measurements. Reproducibility was

enhanced as best as possible by controlling sensor placement and instructions given to the individuals, properly performing calibration procedures before use on each individual, and previous evaluation of the reliability and validity of the system [15, 16].

2.7 | Statistical Analysis

Following confirmation of normal distribution of the data (using Shapiro–Wilk tests), parametric statistical analyses were used. The parameters of malocclusion were evaluated according to their nature. Continuous variables were used for the measurement of overjet and overbite recorded in millimetres. Binary variables were used to assess the presence or absence of specific malocclusions such as anterior and posterior crossbite (coded 1 for presence and 0 for absence). The sagittal molar relationships were transformed into a continuous variable for analytical purposes: 1 for Class II, 2 for Class I and 3 for Class III malocclusion.

Head posture deviations (in the three axes separately; namely pitch, roll and yaw) for the three experimental wheelchair-bound groups were assessed in respect to the number of standard deviations from the norm (the norm and standard deviation being defined by the healthy control group). These head posture measurements were therefore expressed relative to reference population norms as deviations in standard deviations, consistent with what is sometimes carried out in research on rare diseases [17–19]. A one-way ANOVA, with post hoc tests (Tukey and Bonferroni), was conducted to examine whether the head posture axis (pitch, roll and yaw) varied significantly across the different experimental wheelchair-bound groups.

Paired sample *t*-tests were conducted within each wheelchair-bound group to assess whether statistically significant differences existed between the different axes (pitch, roll, yaw) to assess which axis showed more variation in each group. Paired correlations were also assessed to determine whether deviations in one axis were associated with simultaneous deviations in any of the other measured axes.

To explore the relationships between head posture axes (pitch, roll, yaw) and intermaxillary occlusal characteristics (sagittal molar relationships, overjet, overbite and the presence or absence of crossbite), Pearson correlations were carried out on the whole wheelchair-bound experimental sample. All statistical analyses were performed using SPSS (IBM SPSS Statistics for Windows, version 29, Armonk, NY, USA) and a threshold of $p < 0.05$ was considered as statistically significant.

3 | Results

3.1 | Participants

The experimental sample of the study consisted of 20 children, adolescents and young adults aged 6 years and older, 6 of whom were classified into the full control group, 7 in the partial control group and 7 in the no control group (Table 1). The control group was composed of 20 young adults (Table 1).

TABLE 1 | Baseline characteristics of the study population.

Groups	Total	Females	Males	Age range (years)	Mean age (years)
Full head control	6	3	3	9–19	12.5
Partial head control	7	4	3	6–29	15.7
No head control	7	4	3	15–33	19.7
Control group	20	13	7	20–30	23.6

3.2 | Head Posture in Different Groups of Participants

The control group exhibited slight average deviations in head posture from the neutral head position (defined as 0° in pitch, roll and yaw measurements), namely $-0.2^{\circ} \pm 2.1^{\circ}$ for the pitch, $0.4^{\circ} \pm 1.4^{\circ}$ for the roll and $0^{\circ} \pm 2.3^{\circ}$ for the yaw axes (Table 2). Mean head posture deviations for the three groups of wheelchair-bound children with varying degrees of head control are also shown in Table 2, where one can notice very large standard deviations.

When looking at head posture deviations in the three experimental wheelchair-bound groups in respect to the number of standard deviations away from the control group, all wheelchair-bound individuals exhibited higher deviations in head posture from the neutral head position (Table 3; Figure 1).

TABLE 2 | Mean and standard deviations for head posture in all groups in the three axes.

Group	Pitch (mean $\pm$ SD; degrees)	Roll (mean $\pm$ SD; degrees)	Yaw (mean $\pm$ SD; degrees)
Control group	-0.2 ± 2.1	0.4 ± 1.4	0 ± 2.3
Full head control	5.5 ± 15.8	6.5 ± 7.0	5.7 ± 15.2
Partial head control	-17.9 ± 36.5	-20.9 ± 22.7	1.4 ± 34.4
No head control	-10.1 ± 16.2	-27.4 ± 13.6	-11.9 ± 18.1

TABLE 3 | Number of standard deviations (SD) from the control group, in the wheelchair-bound individuals, for head posture in the three axes, expressed as SD units.

Group	Pitch (mean $\pm$ SD)	Roll (mean $\pm$ SD)	Yaw (mean $\pm$ SD)
Full head control	6.3 ± 4.2	5.1 ± 4.4	5.4 ± 3.6
Partial head control	17.6 ± 4.9	16.7 ± 13.9	10.3 ± 10.0
No head control	6.3 ± 6.3	19.6 ± 9.7	6.6 ± 6.5

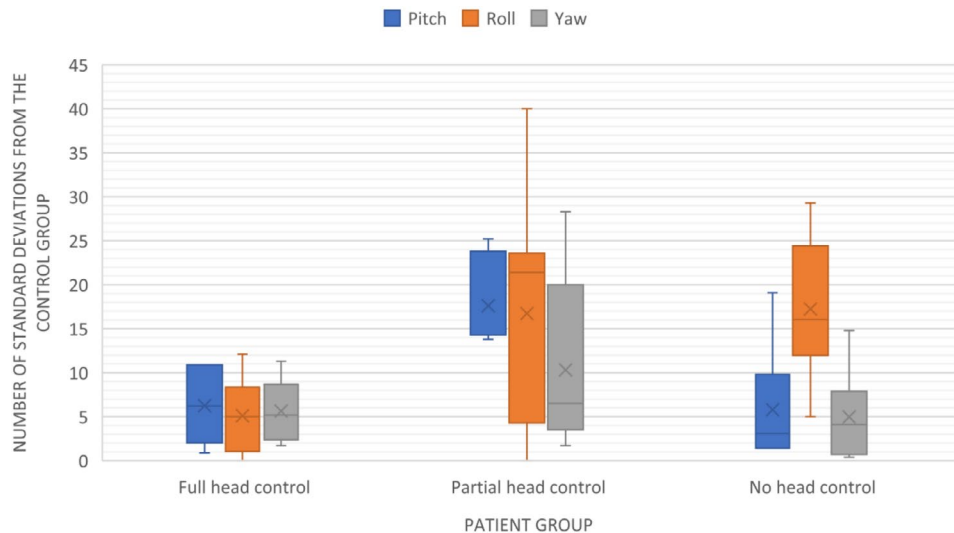


FIGURE 1 | Deviation in head posture (expressed as standard deviation units) for the three groups of wheelchair-bound individuals.

With regard to differences between the control group and the wheelchair-bound groups, participants with full head control exhibited the lowest number of standard deviations from the control group, namely 6.3 for pitch, 5.1 for roll, and 5.6 for yaw. Those with partial head control showed an increased number of standard deviations from the control group, for the pitch (17.6), roll (16.7) and yaw (10.3) axes. The no head control group demonstrated the greatest number of standard deviations from the control group for the roll axis (19.6).

Comparisons between different head control groups in the wheelchair-bound experimental sample showed that the pitch axis was significantly different between the 3 groups ($p=0.001$), whereby participants with partial head control showed significantly more deviation in the pitch axis compared to the other groups (Table 3; Figure 1). The mean difference in the pitch axis between the full head control group and partial control group was -11.4° ($p=0.003$) and between the partial control group and the no control group was 11.3° ($p=0.003$), meaning that participants with partial head control showed significantly more deviation of head posture compared to the full control and no control groups (Table 3; Figure 1). The roll and yaw axes did not show any statistically significant differences between the different wheelchair-bound groups.

When comparing the 3 axes within each wheelchair-bound group, for the full head control group, no statistically significant differences or correlations were found between the pitch, roll and yaw axes. In the partial head control group, a significant paired correlation was found between the pitch and roll axes ($p=0.013$), with deviations in both axes tending to occur simultaneously. In the no head control group, the roll axis was the axis that consistently and significantly showed the greatest amount of deviation when compared to the pitch ($p=0.013$) and yaw ($p=0.015$) axes (Figure 1).

3.3 | Correlation Between Head Posture and Malocclusions

The only deviation in head posture significantly associated with intermaxillary occlusal relationships was deviations in the pitch axis, with good correlation found between pitch deviations and sagittal molar relationships ($r=-0.59$; $p=0.006$).

A backward head posture was correlated with the presence of a Class II malocclusion (with Class II molar relationships), while a forward head posture was correlated with the presence of a Class III malocclusion (with Class III molar relationships) (Table 4).

4 | Discussion

The present study investigated the association between head posture and malocclusions in wheelchair-bound individuals with varying degrees of head control. Our findings indicate that among the three axes of head posture (pitch, roll and yaw), only the pitch axis showed a statistically significant correlation with intermaxillary molar relationships, meaning that those with a more severe backward head posture tended to have Class II molar relationships while those with more severe forward head posture tended to have Class III molar relationships.

These findings are consistent with previous cephalometric studies [7, 12] suggesting that craniofacial growth patterns may be influenced by head positioning, but the present study suggests this for populations with more extreme head posture deviations and using more sophisticated and three-dimensional methodology. Prolonged extension or flexion of the head can possibly affect mandibular positioning and subsequently alter occlusal development during critical growth periods. Additionally, neuromuscular compensation mechanisms may shift mandibular postural position in response to sustained head posture imbalances, thereby influencing sagittal dental relationships [9, 20, 21].

While the roll and yaw axes showed observable deviations in wheelchair-bound individuals with partial or no head control, their correlation with occlusal characteristics was weak and not statistically significant. This implies that the roll and yaw axes may have less of a direct influence on dentoalveolar development compared to the pitch axis. Alternatively, it is possible that their effects on occlusion are secondary to or masked by other factors, such as neuromuscular tone, orofacial function, or compensatory behaviors.

When looking at head posture in wheelchair-bound individuals with varying degrees of head control, it was revealed that

TABLE 4 | Correlations between head posture deviations and intermaxillary occlusal relationships in wheelchair-bound individuals.

Head posture axis	Molar relationships	Overjet	Overbite	Crossbite
Pitch				
Pearson correlation	0.59	0.30	0.14	0.39
Significance (p value)	0.006	0.19	0.56	0.08
Roll				
Pearson correlation	0.26	0.36	-0.17	0.018
Significance (p value)	0.27	0.12	0.94	0.94
Yaw				
Pearson correlation	0.01	0.26	-0.25	0.06
Significance (p value)	0.96	0.27	0.28	0.79

the roll axis differed significantly across groups. Participants with full head control exhibited significantly less lateral head deviation compared to those with partial or no control. While roll itself was not correlated with specific occlusal features, this finding supports the notion that headrest dependency may be associated with substantial postural misalignment.

These results support the notion that altered head posture, particularly in the antero-posterior plane (pitch axis), may influence sagittal molar relationships in wheelchair-bound individuals. Recent research in special-needs populations has highlighted the interplay between postural instability, impaired orofacial muscle coordination and altered mandibular function, suggesting that neuromuscular deficits may amplify the impact of sagittal head deviations on occlusal development [21].

Head posture may thus be one of the contributing factors to the development of malocclusions, especially in individuals with limited head postural control. Clinical assessment of head posture should thus be integrated into orthodontic evaluations for wheelchair-bound individuals with disabilities. Early identification and correction of postural issues could be a preventive strategy to limit the development or progression of malocclusions in this population. Multidisciplinary approaches to optimise body and head alignment through customised wheelchairs and regular postural assessments may represent a non-invasive adjunct to orthodontic treatment.

The primary limitations of the present study were the relatively small sample size, the cross-sectional design, and the heterogeneous aetiologies of the disabilities and ages of the individuals (acting as possible confounders) which limit generalizability and statistical power. As a proof-of-concept study, the finding should be interpreted cautiously, and future studies with larger cohorts are needed to confirm these associations.

Longitudinal studies tracking head posture and occlusal development over time would also be needed in order to establish whether early postural interventions could potentially reduce the incidence or severity of malocclusion in this population.

5 | Conclusions

Important head posture deviations were detected in wheelchair-bound individuals with different levels of head control compared to a sample of seated healthy individuals, with those with no head control showing more extreme deviations.

Significant correlations were found in the present sample between antero-posterior head posture deviations and sagittal molar relationships in wheelchair-bound individuals, whereby a more backward head posture was associated with Class II malocclusions while a more forward posture was associated with Class III malocclusions.

No significant associations were found between the presence of malocclusions and left-right head posture deviations or head tilt deviations.

Author Contributions

Kelly Billiaert: investigation, data curation, formal analysis, writing – original draft. **Mustafa Al-Yassary:** investigation, data curation, formal analysis, writing – original draft. **Stavros Kiliaridis:** conceptualization, methodology; writing – review and editing. **Gregory Antonarakis:** conceptualization, methodology, writing – review and editing; supervision.

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Ethics Statement

This study was approved by the local Research Ethics Committee (Commission cantonale d'éthique de la recherche [CCER], approval number: 2019-00659) and was conducted in accordance with the principles of the Declaration of Helsinki.

Consent

Written informed consent was obtained from all participants or from their legal guardians prior to participation in the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

1. J. Vittek, S. Winik, A. Winik, C. Sioris, A. M. Tarangelo, and M. Chou, "Analysis of Orthodontic Anomalies in Mentally Retarded Developmentally Disabled (MRDD) Persons," *Special Care in Dentistry* 14, no. 5 (1994): 198–202.
2. C. O. Onyeaso, "Orthodontic Treatment Need of Mentally Handicapped Children in Ibadan, Nigeria, According to the Dental Aesthetic Index," *Journal of Dentistry for Children (Chicago, Ill.)* 70, no. 2 (2003): 159–163.
3. K. Winter, L. Baccaglini, and S. Tomar, "A Review of Malocclusion Among Individuals With Mental and Physical Disabilities," *Special Care in Dentistry* 28, no. 1 (2008): 19–26.
4. R. Muppa, P. Bhupathiraju, M. K. Duddu, A. Dandempally, and D. L. Karre, "Prevalence and Determinant Factors of Malocclusion in Population With Special Needs in South India," *Journal of the Indian Society of Pedodontics and Preventive Dentistry* 31, no. 2 (2013): 87–90.
5. R. B. Dinesh, H. M. Arnitha, and A. K. Munshi, "Malocclusion and Orthodontic Treatment Need of Handicapped Individuals in South Canara, India," *International Dental Journal* 53, no. 1 (2003): 13–18.
6. B. Solow and A. Tallgren, "Head Posture and Craniofacial Morphology," *American Journal of Physical Anthropology* 44, no. 3 (1976): 417–435.

7. B. Solow, J. Ovesen, P. W. Nielsen, G. Wildschiodtz, and A. Tallgren, "Head Posture in Obstructive Sleep Apnoea," *European Journal of Orthodontics* 15, no. 2 (1993): 107–114.
8. A. O. Ortega, A. S. Guimaraes, A. L. Ciamponi, and S. K. Marie, "Frequency of Parafunctional Oral Habits in Patients With Cerebral Palsy," *Journal of Oral Rehabilitation* 34, no. 5 (2007): 323–328.
9. V. Martinez-Mihi, V. Paredes-Gallardo, F. J. Silvestre, and J. Silvestre-Rangil, "Comparison of Malocclusion Prevalence, Type and Severity Between Cerebral Palsy and Healthy Subjects: A Prospective Case-Control Study," *Journal of Clinical Medicine* 11, no. 13 (2022): 3711.
10. L. M. Moreno Uribe and S. F. Miller, "Genetics of the Dentofacial Variation in Human Malocclusion," *Orthodontics & Craniofacial Research* 18, no. 1 (2015): 91–99.
11. M. S. Kim, Y. J. Cha, and J. D. Choi, "Correlation Between Forward Head Posture, Respiratory Functions, and Respiratory Accessory Muscles in Young Adults," *Journal of Back and Musculoskeletal Rehabilitation* 30, no. 4 (2017): 711–715.
12. H. Peng, W. Liu, L. Yang, et al., "Does Head and Cervical Posture Correlate to Malocclusion? A Systematic Review and Meta-Analysis," *PLoS One* 17, no. 10 (2022): e0276156.
13. E. von Elm, D. G. Altman, M. Egger, et al., "The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for Reporting Observational Studies," *Journal of Clinical Epidemiology* 61, no. 4 (2008): 344–349.
14. K. Billiaert, M. Al-Yassary, G. S. Antonarakis, and S. Kiliaridis, "Measuring the Difference in Natural Head Position Between the Standing and Sitting Positions Using an Inertial Measurement Unit," *Journal of Oral Rehabilitation* 48, no. 10 (2021): 1144–1149.
15. M. Al-Yassary, K. Billiaert, G. S. Antonarakis, and S. Kiliaridis, "Evaluation of Head Posture Using an Inertial Measurement Unit," *Scientific Reports* 11, no. 1 (2021): 19911.
16. M. Al-Yassary, K. Billiaert, G. S. Antonarakis, and S. Kiliaridis, "Evaluation of Natural Head Position Over Five Minutes: A Comparison Between an Instantaneous and a Five-Minute Analysis With an Inertial Measurement Unit," *Journal of Oral Rehabilitation* 49, no. 4 (2022): 407–413.
17. J. E. Hoover-Fong, K. J. Schulze, A. Y. Alade, et al., "Growth in Achondroplasia Including Stature, Weight, Weight-For-Height and Head Circumference From CLARITY: Achondroplasia Natural History Study—a Multi-Center Retrospective Cohort Study of Achondroplasia in the US," *Orphanet Journal of Rare Diseases* 16, no. 1 (2021): 522.
18. T. Isojima and S. Yokoya, "Development of Disease-Specific Growth Charts in Turner Syndrome and Noonan Syndrome," *Annals of Pediatric Endocrinology and Metabolism* 22, no. 4 (2017): 240–246.
19. U. Mutze, S. F. Garbade, F. Gleich, et al., "Long-Term Anthropometric Development of Individuals With Inherited Metabolic Diseases Identified by Newborn Screening," *Journal of Inherited Metabolic Disease* 46, no. 1 (2023): 15–27.
20. H. Korbmacher, L. Koch, G. Eggers-Stroeder, and B. Kahl-Nieke, "Associations Between Orthopaedic Disturbances and Unilateral Cross-bite in Children With Asymmetry of the Upper Cervical Spine," *European Journal of Orthodontics* 29, no. 1 (2007): 100–104.
21. A. Kimura, V. Z. Drumond, M. G. Oliveira, et al., "Occlusal Features in Individuals With Cerebral Palsy: A Systematic Review and Meta-Analysis," *European Journal of Orthodontics* 47, no. 5 (2025): cjaf070.

DISCUSSION

Cette thèse repose sur une démarche méthodologique en deux étapes, visant à explorer les relations entre posture de la tête et malocclusion chez les individus utilisant un fauteuil roulant comme principal moyen de mobilité. Les deux articles qui la composent combinent validation méthodologique et application clinique.

Le premier article avait pour objectif de valider l'utilisation des unités de mesure inertielle (Inertial Measurement Units, IMU) pour l'évaluation tridimensionnelle de la position naturelle de la tête, en garantissant la fiabilité et la reproductibilité des mesures. Le second article a appliqué cette méthodologie à une population d'individus en fauteuil roulant afin de caractériser la posture de la tête et d'en analyser les relations avec les malocclusions dentaires.

Cette progression méthodologique assure une cohérence scientifique essentielle, en garantissant que les observations cliniques reposent sur un outil de mesure fiable et reproductible.

APPORT METHODOLOGIQUE ET VALIDATION DES IMU

Les résultats du premier article ont mis en évidence une forte concordance entre les mesures issues des IMU et celles obtenues par photographie, en particulier pour l'axe sagittal (pitch), qui constitue l'axe d'analyse pour l'évaluation des relations antéro-postérieures. Cette concordance élevée confirme la validité des IMU pour quantifier l'orientation de la tête de manière précise.

Cette concordance confirme la validité des IMU en tant qu'outil de mesure objectif. Elle est d'autant plus pertinente que les méthodes photographiques, bien que largement utilisées, restent dépendantes de conditions standardisées et d'une analyse indirecte. À l'inverse, les IMU permettent un enregistrement direct et continu de l'orientation de la tête dans les trois plans de l'espace.

Par ailleurs, les analyses de reproductibilité ont montré une excellente fiabilité des mesures, tant en intra-session qu'en inter-session. La faible variabilité observée entre les différentes répétitions de mesures indique que la méthode est peu influencée par les

conditions expérimentales ou par l'opérateur. Ce point est essentiel dans une perspective clinique et de recherche, car il garantit la robustesse des données recueillies.

Ainsi, l'ensemble de ces résultats valide l'utilisation des IMU comme outil fiable, reproductible et pertinent pour l'évaluation de la posture de la tête, constituant une base méthodologique solide pour les analyses réalisées par la suite.

POSTURE DE LA TÊTE EN POSITION ASSISE VERSUS DEBOUT

La comparaison de la posture de la tête entre les positions assise et debout chez des individus sains constitue un élément clé, dans la mesure où elle conditionne la transposabilité des mesures à des populations ne pouvant être évaluées en position standard debout.

Les résultats ont montré l'existence de différences entre ces deux positions selon les axes d'analyse. Dans le plan sagittal (pitch), une tendance à l'extension de la tête a été observée en position assise par rapport à la position debout ($2.5 \pm 2.4^\circ$).

Concernant l'axe frontal (roll) et l'axe transversal (yaw), aucune variation systématique n'a été observée, indiquant une relative stabilité de ces deux axes. Ces résultats suggèrent que l'inclinaison latérale et la rotation de la tête sont peu influencées par le passage de la position debout à la position assise chez des sujets sains.

Dans l'ensemble, bien que des différences aient été observées, celles-ci restent de faible amplitude et ne semblent pas revêtir de signification clinique majeure. Ces résultats soutiennent l'idée que la position assise constitue une alternative valide à la position debout pour l'évaluation de la posture de la tête, ce qui est fondamental pour l'étude de populations spécifiques telles que les individus en fauteuil roulant.

POSTURE DE LA TÊTE CHEZ LES PERSONNES EN FAUTEUIL ROULANT

L'analyse de la posture de la tête chez les individus en fauteuil roulant met en évidence des différences notables par rapport aux sujets sains. Les résultats montrent une variabilité accrue de l'orientation de la tête sur les trois axes, traduisant une instabilité posturale plus importante dans cette population.

Cette variabilité peut être expliquée par plusieurs facteurs, notamment les limitations neuromusculaires, le contrôle de la posture de la tête réduit et la dépendance à des dispositifs de soutien. Les individus présentant un contrôle de la tête diminué ou nécessitant un appui-tête montrent des déviations plus marquées, ce qui souligne l'influence du niveau de contrôle moteur sur la posture de la tête.

Dans le plan frontal (roll), les résultats montrent la plus grande dispersion des valeurs, indiquant que les déviations latérales de la tête constituent l'altération posturale la plus marquée dans cette population. Dans le plan sagittal (pitch), une variabilité accrue est également observée par rapport aux sujets sains, bien que d'amplitude moindre que dans le plan frontal. À l'inverse, dans le plan transversal (yaw), les résultats mettent en évidence une variabilité plus limitée et sans orientation préférentielle. Comparativement aux plans frontal et sagittal, la rotation de la tête apparaît moins affectée, suggérant un meilleur contrôle relatif dans ce plan.

Ces résultats mettent en évidence que la posture de la tête chez les individus en fauteuil roulant ne correspond pas à une position de référence physiologique, mais plutôt à une adaptation fonctionnelle aux contraintes biomécaniques et neuromusculaires propres à cette population.

RELATIONS ENTRE POSTURE DE LA TÊTE ET MALOCCLUSIONS

L'analyse des relations entre posture de la tête et malocclusions a permis de mettre en évidence des associations spécifiques selon les axes d'analyse. Dans le plan sagittal (pitch), une relation significative a été observée entre l'orientation de la tête et les classes molaires. Une posture en extension est plus fréquemment associée à des malocclusions de Classe II, tandis qu'une posture en flexion est observée chez les individus présentant des relations de Classe III.

Ces résultats sont cohérents avec les modèles fonctionnels de croissance cranio-faciale. L'orientation céphalique influence la position mandibulaire et l'équilibre musculaire cervico-facial. Une extension céphalique est associée à une rotation mandibulaire postérieure et à une augmentation de la dimension verticale antérieure, caractéristiques des Classes II. À l'inverse, une flexion céphalique favorise une rotation mandibulaire antérieure, compatible avec les Classes III[7, 11, 12].

En revanche, aucune association significative n'a été mise en évidence dans les plans frontal (roll) et transversal (yaw), ce qui suggère que l'influence de la posture de la tête sur les malocclusions est principalement limitée au plan sagittal.

Ces résultats apportent une confirmation tridimensionnelle et objective d'observations déjà suggérées dans la littérature, tout en renforçant la compréhension des interactions entre posture et morphologie cranio-faciale.

IMPLICATIONS CLINIQUES

Les résultats de ces études présentent plusieurs implications cliniques importantes. L'utilisation des IMU permet une évaluation objective, reproductible et tridimensionnelle de la posture de la tête, offrant ainsi un outil complémentaire pertinent à l'analyse céphalométrique traditionnelle.

L'intégration de l'analyse posturale dans le diagnostic orthodontique pourrait améliorer la compréhension des déséquilibres fonctionnels associés aux malocclusions, en particulier chez les patients présentant des troubles neuromusculaires ou des limitations fonctionnelles.

Chez les individus en fauteuil roulant, la prise en compte de la posture de la tête et des conditions d'installation pourrait contribuer à une approche thérapeutique plus globale. L'optimisation du positionnement, notamment à travers les dispositifs de soutien céphalique, pourrait avoir un impact sur l'équilibre oro-facial et potentiellement sur l'évolution des malocclusions. Ces éléments soulignent l'intérêt d'une approche multidisciplinaire pour ces personnes.

LIMITATIONS

Certaines limites doivent être prises en compte dans l'interprétation des résultats. La taille de l'échantillon, notamment dans la population en fauteuil roulant, reste limitée, ce qui peut réduire la puissance statistique et la généralisation des résultats.

L'hétérogénéité des pathologies sous-jacentes et des niveaux de contrôle de la position de la tête constitue également une source de variabilité importante. Par ailleurs, bien que les IMU permettent une mesure précise de l'orientation de la tête, elles ne fournissent pas d'information directe sur les structures squelettiques.

Enfin, le design transversal de l'étude ne permet pas d'établir de relation causale entre posture céphalique et malocclusions, limitant l'interprétation des résultats à des associations.

PERSPECTIVES DE RECHERCHE

Des études longitudinales seraient nécessaires afin d'examiner l'évolution de la posture de la tête au cours de la croissance et son impact potentiel sur le développement cranio-facial. L'enregistrement prolongé des données posturales pourrait notamment permettre d'identifier des profils dynamiques spécifiques associés à certaines malocclusions.

À plus long terme, l'intégration de mesures objectives de la posture dans l'analyse orthodontique pourrait contribuer au développement d'approches diagnostiques et thérapeutiques plus personnalisées, fondées sur une compréhension plus globale des interactions entre posture, fonction neuromusculaire et morphologie cranio-faciale.

CONCLUSION

Cette thèse avait pour objectif d'évaluer le rôle potentiel de la posture de la tête dans l'apparition des malocclusions, en s'appuyant sur une approche méthodologique innovante fondée sur l'utilisation des unités de mesure inertielle (IMU).

La validation préalable des IMU a démontré leur fiabilité et leur pertinence pour l'analyse tridimensionnelle de la position naturelle de la tête, y compris en position assise. Cette étape a permis d'établir une base méthodologique solide pour l'étude clinique menée chez des individus en fauteuil roulant.

L'application de cette méthodologie a mis en évidence des variations significatives de l'orientation de la tête, ainsi qu'une association entre posture antéro-postérieure de la tête et relations molaires sagittales. Ces résultats suggèrent que la posture de la tête pourrait constituer un facteur fonctionnel impliqué dans l'équilibre occlusal, en particulier dans des populations présentant des contraintes neuromusculaires spécifiques.

Ces résultats ouvrent ainsi la voie à des investigations longitudinales visant à préciser l'influence des déterminants posturaux sur le développement cranio-facial.

REFERENCES

1. Vittek J, Winik S, Winik A, Sioris C, Tarangelo AM, Chou M: **Analysis of orthodontic anomalies in mentally retarded developmentally disabled (MRDD) persons.** *Spec Care Dentist* 1994, **14**(5):198-202.
2. Muppa R, Bhupathiraju P, Duddu MK, Dandempally A, Karre DL: **Prevalence and determinant factors of malocclusion in population with special needs in South India.** *J Indian Soc Pedod Prev Dent* 2013, **31**(2):87-90.
3. Onyeaso CO: **Orthodontic treatment need of mentally handicapped children in Ibadan, Nigeria, according to the dental aesthetic index.** *J Dent Child (Chic)* 2003, **70**(2):159-163.
4. Winter K, Baccaglini L, Tomar S: **A review of malocclusion among individuals with mental and physical disabilities.** *Spec Care Dentist* 2008, **28**(1):19-26.
5. Moreno Uribe LM, Miller SF: **Genetics of the dentofacial variation in human malocclusion.** *Orthod Craniofac Res* 2015, **18** Suppl 1(0 1):91-99.
6. Solow B, Tallgren A: **Head posture and craniofacial morphology.** *Am J Phys Anthropol* 1976, **44**(3):417-435.
7. Solow B, Ovesen J, Nielsen PW, Wildschiodtz G, Tallgren A: **Head posture in obstructive sleep apnoea.** *Eur J Orthod* 1993, **15**(2):107-114.
8. Ortega AO, Guimaraes AS, Ciamponi AL, Marie SK: **Frequency of parafunctional oral habits in patients with cerebral palsy.** *J Oral Rehabil* 2007, **34**(5):323-328.
9. Martinez-Mihi V, Paredes-Gallardo V, Silvestre FJ, Silvestre-Rangil J: **Comparison of Malocclusion Prevalence, Type and Severity between Cerebral Palsy and Healthy Subjects: A Prospective Case-Control Study.** *J Clin Med* 2022, **11**(13).
10. Huard P: **Paul Broca (1824-1880). avec une bibliographie des travaux de Broca par Samuel Pozzi (1846-1918).** *Revue d'histoire des sciences* 1961:47-86.
11. Kim MS, Cha YJ, Choi JD: **Correlation between forward head posture, respiratory functions, and respiratory accessory muscles in young adults.** *J Back Musculoskeletal Rehabil* 2017, **30**(4):711-715.
12. Peng H, Liu W, Yang L, Zhong W, Yin Y, Gao X, Song J: **Does head and cervical posture correlate to malocclusion? A systematic review and meta-analysis.** *PLoS One* 2022, **17**(10):e0276156.
13. Solow B, Tallgren A: **Natural head position in standing subjects.** *Acta Odontol Scand* 1971, **29**(5):591-607.
14. Audette I, Dumas JP, Cote JN, De Serres SJ: **Validity and between-day reliability of the cervical range of motion (CROM) device.** *J Orthop Sports Phys Ther* 2010, **40**(5):318-323.
15. Bass NM: **The aesthetic analysis of the face.** *European Journal of Orthodontics* 1991, **13**(5):343-350.
16. Lundström A, Forsberg C-M, Westergren H, Lundström F: **A comparison between estimated and registered natural head posture.** *European Journal of Orthodontics* 1991, **13**(1):59-64.
17. Lundström F, Lundström A: **Natural head position as a basis for cephalometric analysis.** *Am J Orthod Dentofacial Orthop* 1992, **101**(3):244-247.
18. Patel S, Park H, Bonato P, Chan L, Rodgers M: **A review of wearable sensors and systems with application in rehabilitation.** *J Neuroeng Rehabil* 2012, **9**:21.
19. Ghislieri M, Gastaldi L, Pastorelli S, Tadano S, Agostini V: **Wearable Inertial Sensors to Assess Standing Balance: A Systematic Review.** *Sensors (Basel)* 2019, **19**(19).

20. Leung MY, Lo J, Leung YY: **Accuracy of Different Modalities to Record Natural Head Position in 3 Dimensions: A Systematic Review.** *J Oral Maxillofac Surg* 2016, **74**(11):2261-2284.
21. Al-Yassary M, Billiaert K, Antonarakis GS, Kiliaridis S: **Evaluation of head posture using an inertial measurement unit.** *Sci Rep* 2021, **11**(1):19911.
22. Cuccia AM, Lotti M, Caradonna D: **Oral breathing and head posture.** *Angle Orthod* 2008, **78**(1):77-82.
23. Neiva PD, Kirkwood RN, Mendes PL, Zabjek K, Becker HG, Mathur S: **Postural disorders in mouth breathing children: a systematic review.** *Braz J Phys Ther* 2018, **22**(1):7-19.
24. Bjehin R: **A Comparison Between the Frankfort Horizontal and the Sella Turcica - Nasion as Reference Planes in Cephalometric Analysis.** *Acta Odontologica Scandinavica* 1957, **15**(1):1-12.
25. Moorrees CFA, Kean MR: **Natural head position, a basic consideration in the interpretation of cephalometric radiographs.** *American Journal of Physical Anthropology* 1958, **16**:213-234.
26. Huggare JA: **A natural head position technique for radiographic cephalometry.** *Dentomaxillofac Radiol* 1993, **22**(2):74-76.